Zhirui Dai

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Research Interests

My research pursues spatiotemporal memory for embodied agents: real-time, scalable, dynamic 3D scene representations that unify multiple modalities, such as geometry (signed distance and occupancy), appearance (color) and open-vocabulary semantics, into a single memory that robots can query for safe motion planning, navigation, mobile manipulation and task planning. A recurring theme of my work is making these methods fast enough to run onboard, through C++, custom CUDA kernels and open-source software.

Education

University of California San Diego — **San Diego, CA, USA**
Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Nikolay Atanasov — *01/2022 - Expected 06/2027*

University of California San Diego — **San Diego, CA, USA**
M.S. in Electrical and Computer Engineering; Research with Prof. Nuno Vasconcelos — *09/2019 - 06/2021*

Fudan University — **Shanghai, China**
B.S. in Physics — *09/2015 - 06/2019*

Honors & Awards

Best Paper Award, RSS 2025 Workshop on Leveraging Implicit Methods for Aerial Autonomy 2025
Best Paper Nomination, RSS 2025 Workshops on Mobile Manipulation & Learned Robot Representations 2025
Electrical and Computer Engineering Department Fellowship, UCSD 2022
Outstanding Graduate, Fudan University 2019

Publications

Underline: this author; *: equal contribution. Google Scholar: profile.

Journal & Conference Papers

- Z. Dai*, T. Fan*, M. Amani, J. Seo, K. M. B. Lee, H. Oh, N. Atanasov. "**Kernel-SDF: An Open-Source Library for Real-Time Signed Distance Function Estimation using Kernel Regression.**" IEEE Robotics and Automation Letters (RA-L), 2026. Paper Project
- Z. Dai*, Q. Qian*, T. Fan, N. Atanasov. "**OREN: Octree Residual Network for Real-Time Euclidean Signed Distance Mapping.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Paper Project
- T. Matthews, M. Kumar A. R., A. Begzadić, S. Tonkens, Z. Dai, N. Atanasov, J. Cortés, S. Herbert. "**Expanding Safe Sets with Learning-Based Barrier Functions and Reachability.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Paper
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. "**Learning Scene-Level Signed Directional Distance Function with Ellipsoidal Priors and Neural Residuals.**" IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. Paper Project
- S. Kim, W. Chung, Z. Dai, D. Bhatt, A. Shukla, H. Su, Y. Tian, N. Atanasov. "**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**" IEEE International Conference on Robotics and Automation (ICRA), 2026. Paper Project
- B. Rabiei, M. Kumar A. R., Z. Dai, S. L. S. R. Pilla, Q. Dong, N. Atanasov. "**LTLCodeGen: Code Generation of Syntactically Correct Temporal Logic for Robot Task Planning.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. Paper Project
- K. Long, Y. Yi, Z. Dai, S. Herbert, J. Cortés, N. Atanasov. "**Sensor-Based Distributionally Robust Control for Safe Robot Navigation in Dynamic Environments.**" The International Journal of Robotics Research (IJRR), 2025. Paper Project
- Z. Dai, A. Asgharivaskasi, T. Duong, S. Lin, M.-E. Tzes, G. J. Pappas, N. Atanasov. "**Optimal Scene Graph Planning with Large Language Model Guidance.**" IEEE International Conference on Robotics and Automation (ICRA), 2024. Paper Project
- Z. Dai, Y. Jiang, Y. Li, B. Liu, A. B. Chan, N. Vasconcelos. "**BEV-Net: Assessing Social Distancing Compliance by Joint People Localization and Geometric Reasoning.**" IEEE/CVF International Conference on Computer Vision (ICCV), 2021. Paper

Preprints

- M. Amani*, M. Alyaseen*, Z. Dai, K. M. B. Lee, N. Atanasov, J. Cortés. "**BubbleCBF: Constructive Safety for Second-Order Systems and Application to Autonomous Robot Navigation in Unknown Environments.**" Under review, 2026. Paper Project
- Z. Dai, Q. Qian, D. M. Nguyen, Q.-D. Pham, K. Bronder, C. Nieto-Granda, Y. Chen, Q. Nguyen, N. Atanasov. "**OREN-X: Octree Residual Network for Real-Time Multi-Modal Mapping.**" Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. Paper Project
- Q.-D. Pham, A. Dao, D. V. Le, N. V. T. Pham, T. A. Nguyen, Z. Dai, Y. Chen, T. P. Le, V.-T. Nguyen, Q. Nguyen. "**AdaGeoVLN: Selective Geometry Across Representation Depth and Navigation Time for Vision-Language Navigation.**" Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. Paper
- J. Stanley*, Z. Dai*, Q. Qian, T.-C. Ho, T. Fan, S. Saha, C. Barngrover, K. M. B. Lee, N. Atanasov. "**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**" Submitted to Autonomous Robots, 2026, under review. Paper Project
- K. M. B. Lee, Z. Dai, C. Le Gentil, L. Wu, N. Atanasov, T. Vidal-Calleja. "**Safe Bubble Cover for Motion Planning on Distance Fields.**" arXiv preprint arXiv:2408.13377, 2024. Paper

Workshop Papers

- J. Stanley*, Z. Dai*, T.-C. Ho, S. Saha, C. Barngrover, N. Atanasov. "**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**" ICRA 2026 Workshop on Aerial Inspection for Marine Infrastructures (AIMI), 2026. Paper Project
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. "**Learning Scene-Level Signed Directional Distance Function for Aerial Autonomy.**" Workshop on Leveraging Implicit Methods for Aerial Autonomy at RSS, 2025. Paper Project **[Best Paper Award]**
- S. Kim, W. Chung, Y. Tian, Z. Dai, A. Shukla, H. Su, N. Atanasov. "**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**" Workshop on Mobile Manipulation and Workshop on Learned Robot Representations at RSS, 2025. Paper Project **[Best Paper Nomination]**

Research Experience

Research Intern, VinMotion Inc. **Irvine, CA, USA**
Real-Time Multi-Modal 3D Mapping (in collaboration with ERL, UCSD) *06/2026 - 09/2026*

- **OREN-X: Real-Time Multi-Modal Mapping** *Submitted to ICRA 2027*
  - Developed a unified octree representation that jointly learns SDF, occupancy, radiance and vision-language fields online, with cross-modality supervision and online dictionary learning (3.7x feature compression); 80+ fps for SDF and 30+ fps for all modalities.
  - Improved near-surface SDF accuracy by 33% (over OREN) and open-vocabulary 3D mIoU by 71%; validated on Replica, TUM RGB-D, a Jackal robot and a humanoid across different scene scales (studio, lab space and construction site).

Graduate Student Researcher, Existential Robotics Lab, UCSD **San Diego, CA, USA**
Advisor: Prof. Nikolay Atanasov *01/2022 - Present*

- **Distance-Field Representations for Mapping & Planning** *TPAMI 2026, IROS 2026, RA-L 2026*
  - Introduced neural signed directional distance functions (SDDF) with ellipsoidal priors and neural residuals, enabling single-pass occlusion reasoning and collision checking with SOTA accuracy on Replica.
  - Developed OREN, an octree residual network for globally accurate real-time SDF learning from LiDAR/RGB-D (7 fps on Jetson Orin).
  - Developed Kernel-SDF for SDF estimation with uncertainty quantification, and distance-accelerated UAV motion planning (Bubble*) deployed onboard a quadrotor.
- **3D Vision-Language Maps for Mobile Manipulation** *ICRA 2026*
  - Co-developed SBP, a 3D latent map of VLM features that serves as spatiotemporal memory for a transformer manipulation policy, improving success by up to 15% over image-based policies.
- **Language-Driven Task Planning with Guarantees** *ICRA 2024, IROS 2025*
  - Combined LLM guidance with consistent LTL heuristics for optimal scene graph planning (1600x speedup over AMRA*), and developed LTLCodeGen for syntactically correct LTL generation.

Graduate Student Researcher, Statistical Visual Computing Lab, UCSD **San Diego, CA, USA**
Advisor: Prof. Nuno Vasconcelos *06/2020 - 08/2021*

- **Geometric Reasoning for Social Distance Monitoring** *ICCV 2021*
  - Proposed BEV-Net, jointly localizing people and reasoning about geometry via a bird's-eye-view projection layer with calibration-free camera pose recovery.

Teaching & Mentoring

Teaching Assistant, ECE276B: Planning & Learning in Robotics, UCSD — Spring 2023, Spring 2024
Research mentor, Qihao Qian (M.S.): co-first author of OREN (IROS 2026); co-author of OREN-X — 02/2025 - Present
Research mentor, Tianxing Fan (M.S.): co-first author of Kernel-SDF (RA-L 2026); co-author of OREN — 12/2024 - 02/2026
Mentor, ENLACE, SRIP and STARS Summer Research Programs, UCSD — 07/2022 - 08/2022
Mentor, Guided Engineering Apprenticeship in Research (GEAR), UCSD — 09/2021 - 07/2022

Professional Service

Conference Reviewer: NeurIPS, CVPR, RSS, CoRL, ICRA, IROS
Journal Reviewer: IEEE Robotics and Automation Letters (RA-L), IEEE Trans. on Intelligent Transportation Systems (T-ITS)

Open-Source Software

Released: erl_geometry (C++/CUDA semi-sparse octree), Kernel-SDF (C++/Python/ROS), neural_sddf (PyTorch + CUDA ops), BEV-Net
Contributions to: PyTorch, Open3D, PointCloudLibrary, pybind11
Skills: C++, CUDA, Python, PyTorch, ROS, OpenCV, Open3D, Docker, Kubernetes, ManiSkill

Industry Experience

Research Intern, XCOM Labs Inc. (neural pose prediction for VR/AR remote rendering) — 08/2021 - 12/2021
Software Development Intern, Camel Microelectronics Inc. (SoC IDE & embedded toolchain) — 01/2018 - 06/2019